brings with itself most of the ocean bottoms' organic material. Suddenly exposed to the atmosphere, this material could decay, converting atmospheric O_2 into lethal amounts of CO_2.

Conceivable, yes. But highly improbable on two counts: highly unlikely that there is 1000 times more undecayed ocean-bottom organic material than geophysicists think, and highly unlikely that a sufficiently vigorous oceanic overturn will occur.[25]

Nevertheless, in *Interstellar* the Earth is surely dying and humanity must find a new home. The solar system, aside from Earth, is inhospitable, so the search is on, beyond our solar system.

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24 Chloroplasts and photosynthesis also occur in algae, and in cyanobacteria in the ocean, both of which I treat as plant life in my simplified description. (In some sense, cyanobacteria are a form of chloroplast.)

25 For some quantitative details and explanations of the huge uncertainties in the geophysical estimates, see *Some Technical Notes* at the end of the book.